

# On the Role of the Diffusion Matrix in Stiffness Mitigation for Stochastic Particle Flow Filters

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**Abstract**—Stochastic particle flow filters are driven by diffusion processes. Recently it has been discovered that a stochastic particle flow filter exists for any given diffusion matrix. Therefore, the diffusion matrix effectively acts as a design parameter. A variety of diffusion matrices have been proposed in the literature. One natural question is: How to select the diffusion matrix to improve filtering performance? Particle flow filters are described by stochastic differential equations and realized through numerical discretization whose accuracy is affected by both the stiffness of the differential equations and the value of the diffusion matrix. In this paper, we examine the role of the diffusion matrix in stiffness mitigation of stochastic particle flows and its trade-off with numerical accuracy. Objective functions are proposed to balance several conflicting factors of stiffness mitigation and numerical accuracy. Analytic solutions of optimal diffusion matrices are derived for stiffness mitigation of stochastic particle flows. Results are illustrated using numerical examples.

**Index Terms**—particle flow filters, nonlinear filtering, stiffness mitigation, condition number

## I. INTRODUCTION

Various particle flow filters have been proposed for density based nonlinear filtering [8]. Some flows are deterministic while others are stochastic. Implementations of particle flow filters in a wide range of applications have indicated that stochastic particle flows lead to robust performance [1], [2], [9], [11], [13], [15], [17], [20], [21], [22], [23], [24], [26], [28], [29], [31], [33]. Several stochastic flows with different diffusion matrices have been proposed in the literature. One natural question is what are the impacts of different diffusion matrices on filtering performance? What is the optimal choice of the diffusion matrix?

To answer these questions, we revisit the design of particle flow filters and examine the role of the diffusion matrix in stiffness mitigation. Stochastic particle flows are represented as solutions to stochastic differential equations which are realized in applications by difference equations through discretization. Stiff differential equations pose challenges to obtaining numerical approximations of exact theoretical solutions with guaranteed accuracy [4], [12], [18]. While it has not been possible to define stiffness in precise mathematical terms in a satisfactory manner, [18] refers stiffness as a general phenomenon in which a numerical method is “forced to use a step-length that is excessively small in relation to the smoothness of the exact solution”. Stiffness mitigation in particle flow filters has been discussed in the literature for a given diffusion matrix. A

number of techniques for stiffness mitigation were discussed in [7]. An approach of adaptive step size in the discretization of particle flows to increase numerical accuracy was proposed in [19]. Numerical integration of the closed form of linear flows to avoid stiffness was discussed in [3]. In those works, the diffusion matrix was assumed to be given and fixed.

In this paper, we investigate the role of the diffusion matrix in stiffness mitigation of particle flows. It turns out that properly increasing the value of the diffusion matrix, i.e., its norm, may reduce the stiffness of the underlying differential equations of the flow. One complication is that an increase in the diffusion matrix may increase the variance of the corresponding difference equation, which would lead to reduced filtering accuracy without increasing the number of particles. Another consideration is computational overhead in obtaining a suitable diffusion matrix. We formulate the problem of finding an appropriate diffusion matrix as an optimization problem by minimizing an objective function that provides a trade-off between stiffness mitigation and numerical accuracy. The computation overhead is addressed by obtaining analytical solutions. Two numerical examples are included to illustrate the results, which indicates that the overhead is negligible.

A filter, including a particle flow filter, is constructed for sequential state estimation, and its state estimation is updated as new data is collected. To keep notations as simple as possible and without loss of generality, we are mainly concerned with one-step Bayesian estimations. In practice, its implementation is sequential consisting of multiple one-step estimations. Another consideration is that the framework is applicable to general Bayesian estimation/inference such as those in machine learning which are different from filtering problems.

The rest of the paper is organized as the following. In Section 2, we introduce a parameterized form of stochastic particle flow filters in which the diffusion matrix is a design parameter. In Section 3, we investigate the exploitation of the freedom in the choice of the diffusion matrix for stiffness mitigation of the underlying stochastic differential equations. Several methods are proposed for choosing an appropriate diffusion matrix. Finally, the results are illustrated using numerical examples in Section 4.

## II. A PARAMETERIZED FORM OF STOCHASTIC PARTICLE FLOWS

Assume a given probability space on which a continuous-time stochastic system is defined. The state vector of the system is represented by  $\{x_t \in \mathbb{R}^n, t \in [t_0, T]\}$  in which  $x_t$  is the state vector at time  $t$  and  $[t_0, T]$  is time interval of interest. A number of measurements as functions of the state vector  $x_t$  are taken at discrete time instants  $\{t_k, k = 1, 2, \dots\}$ . At  $t_k$ , the measurement  $z_k \in \mathbb{R}^d, k = 1, 2, \dots$ . The collection of measurements up to time  $t_k$  is denoted as  $Z_k = \{z_i, i = 1, 2, \dots, k\}$ . According to Bayes's Theorem, the distribution of the state vector immediately after measurement at  $t_k$  is updated as [14]

$$p(x, t_k | Z_k) = \frac{p(z_k | x, t_k) p(x, t_k | Z_{k-1})}{p(z_k | Z_{k-1})} \quad (1)$$

in which  $p(z_k | x, t_k)$  is the probability density (likelihood) of measurement  $z_k$  at time  $t_k$  conditioned on  $x$ ,  $p(x, t_k | Z_{k-1})$  is the probability density of prior distribution of  $x$ , and  $p(x, t_k | Z_k)$  is the probability density of the posterior distribution. Without loss of generality, it is assumed throughout this paper that all probability density functions exist, sufficiently (second order) differentiable, and are non-vanishing everywhere. Otherwise we restrict discussions to the supports of the density functions.

For simplicity, we denote

$$\begin{aligned} p(x) &= p(x, t_k | Z_k), & g(x) &= p(x, t_k | Z_{k-1}), \\ h(x) &= p(z_k | x, t_k), & p_z(z) &= p(z_k | Z_{k-1}). \end{aligned}$$

Then (1) becomes

$$p(x) = \frac{g(x)h(x)}{p_z(z)},$$

where  $p_z(z)$  is also known as the normalization factor. In the framework of particle flow filters, particle flows are defined through homotopy. Toward that end, we define a new conditional probability density function as the following

$$p(x, \lambda) = \frac{g(x)h^\lambda(x)}{c(\lambda)} \quad (2)$$

for all  $\lambda \in [0, 1]$ . In (2),  $c(\lambda)$  is the normalization factor so that  $p(x, \lambda)$  remains a probability density function for all  $\lambda \in [0, 1]$ . It's clear from (2) that

$$p(x, 0) = g(x), \quad p(x, 1) = p(x).$$

In other words,  $p(x, 0)$  is the density function of the prior distribution and  $p(x, 1)$  is that of the posterior distribution. The mapping  $p(x, \lambda) : \mathbb{R}^+ \times [0, 1] \rightarrow \mathbb{R}^+$  in (2) defines a homotopy from  $g(x)$  to  $p(x)$ . By taking the natural logarithm on both sides of (2), we obtain

$$\log p(x, \lambda) = \log g(x) + \lambda \log h(x) - \log c(\lambda). \quad (3)$$

Therefore,  $\log p(x, \lambda)$  is a  $\lambda$ -varying linear combination of  $\log g(x)$  and  $\log h(x)$ . The concept of particle flow filtering states that we may construct an artificial random process  $\{\alpha(\lambda), \lambda \in [0, 1]\}$  such that the probability density function

of  $\alpha(\lambda)$  is exactly  $p(x, \lambda)$  for all  $\lambda \in [0, 1]$ . If this is possible, the density functions of  $\alpha(0)$ ,  $\alpha(1)$  would be the prior  $pg(x)$  and the posterior  $p(x)$ , respectively.

It turns out that there is much freedom in the construction of  $\{\alpha(\lambda), \lambda \in [0, 1]\}$  [8]. The  $\alpha(\lambda)$  could be driven by a deterministic process as in the Exact Flow [8], or by a stochastic process as in stochastic flows [6]. In this paper, we focus on stochastic flows in which  $\alpha(\lambda)$  is driven by a diffusion process as the following

$$d\alpha = f(\alpha, \lambda)d\lambda + q(\alpha, \lambda)dw_\lambda \quad (4)$$

where  $f(\alpha, \lambda) \in \mathbb{R}^n$  is a drift function,  $q(\alpha, \lambda) \in \mathbb{R}^{n \times m}$  is a diffusion matrix, and  $w_\lambda \in \mathbb{R}^m$  is an  $m$  dimensional Brownian motion in  $\lambda$  with  $E[dw_\lambda dw_\lambda^T] = \sigma(\lambda)d\lambda$ . The stochastic differential equation (4) defines a standard diffusion process [14]. Note that  $\{\alpha(\lambda), \lambda \in [0, 1]\}$  is a stochastic process in  $\lambda$ , not in time. For clarity, we drop its dependence on  $\lambda$  but add the dependence back when it is beneficial to emphasize its dependence on  $\lambda$ . Without loss of generality, we assume that  $\sigma(\lambda) = I_m$ , an  $m \times m$  identity matrix, and denote

$$Q(\alpha, \lambda) = q(\alpha, \lambda)q(\alpha, \lambda)^T \in \mathbb{R}^{n \times n}.$$

The matrix  $Q(\alpha, \lambda) = [Q_{i,j}]$  is always symmetric positive semi-definite for any  $\alpha$  and  $\lambda$ .

The design of particle flow filters is to choose the drift function  $f$  and the diffusion matrix  $q$ , or equivalently  $Q$ , such that the density function of  $\alpha(\lambda)$  is  $p(x, \lambda)$  for all  $\lambda \in [0, 1]$ . Recently, a parameterized family of stochastic flows was derived in [5] for any diffusion matrix  $q(\lambda)$  that is not a function of  $\alpha$ , as the following:

$$f = S^{-1}[-\nabla_x \log h + KS^{-1}\nabla_x \log p]|_{x=\alpha}, \quad (5)$$

$$K = \frac{1}{2}SQ(\lambda)S + \frac{1}{2}(\nabla_x \nabla_x^T \log h), \quad (6)$$

in which  $S \triangleq \nabla_x \nabla_x^T \log p$  is assumed nonsingular for all  $\lambda \in [0, 1]$ .

This new family of particle flows takes several particle flows existing in the literature as special cases. The matrix  $Q$  serves as a design parameter a user may choose. For convenience, we re-write (5) as

$$f(\alpha, \lambda) = f_1(\alpha, \lambda) + Qf_2(\alpha, \lambda) \quad (7)$$

in which

$$\begin{aligned} f_1(\alpha, \lambda) &= -S^{-1}\nabla_x \log h|_{x=\alpha} \\ &+ \frac{1}{2}S^{-1}(\nabla_x \nabla_x^T \log h)S^{-1}\nabla_x \log p|_{x=\alpha}, \end{aligned} \quad (8)$$

and

$$f_2(\alpha, \lambda) = \frac{1}{2}\nabla_x \log p|_{x=\alpha}. \quad (9)$$

Note that  $f_1$  and  $f_2$  are not functions of  $Q$ . The form (7) is linear in  $Q$ .

### III. STIFFNESS MITIGATION

When particle flow filters are implemented in practice, we typically generate a number of, say  $N$ , particles with initial conditions  $\{\alpha_i(0), i = 1, 2, \dots, N\}$ , propagate the particles to  $\{\alpha_i(1), i = 1, 2, \dots, N\}$  by numerically evaluating the flow equation (4), and construct an estimate(s) using  $\{\alpha_i(1), i = 1, 2, \dots, N\}$ . For example, the posterior mean of  $x$  may be estimated as the average of  $N$  particles:

$$\hat{x} = \frac{1}{N} \sum_i \alpha_i(1),$$

and the covariance of the estimate may be estimated as the covariance of the particles  $\{\alpha_i(1)\}$ . Therefore, numerically evaluating the flow equation (4) with sufficient accuracy is important to the implementation of particle flow filters. In this section, we investigate the exploitation of the freedom in the choice of the diffusion matrix  $q$  (or equivalently  $Q$ ) to improve the numerical evaluation of the stochastic flow equation (4).

It has been long known that certain deterministic or stochastic differential equations could be stiff [4], [12], [18], which poses challenges for their numerical evaluation. While the phenomenon of stiffness has been widely observed, a precise mathematical definition of stiffness has not been available. One approach of linking stiffness to a mathematical formulation is to use the Jacobian matrix to describe the degree of stiffness of a differential equation, known as the linear stability theory [18]. Consider an abstract  $n$ -dimensional ordinary differential equation

$$\frac{dx(t)}{dt} = f(x). \quad (10)$$

Without loss of generality, assume that  $f(0) = 0$ . Denote

$$F = \nabla_x f|_{x=0}.$$

Assume that all eigenvalues of  $F$  have non-positive real parts and some have strictly negative real parts. Let  $\{\lambda_i, i = 1, 2, \dots, n\}$  be the collection of the eigenvalues of  $F$  and define  $\bar{\lambda}, \underline{\lambda} \in \{\lambda_i, i = 1, 2, \dots, n\}$  such that

$$|\operatorname{Re} \bar{\lambda}| = \max_i |\operatorname{Re} \lambda_i|, \quad |\operatorname{Re} \underline{\lambda}| = \min_i |\operatorname{Re} \lambda_i|. \quad (11)$$

The stiffness ratio of  $F$  is defined as [18]

$$R_{\text{stiff}} = \begin{cases} \frac{|\operatorname{Re} \bar{\lambda}|}{|\operatorname{Re} \underline{\lambda}|}, & \text{if } |\operatorname{Re} \underline{\lambda}| > 0, \\ +\infty, & \text{if } |\operatorname{Re} \underline{\lambda}| = 0. \end{cases}$$

The ordinary differential equation (10) is stiff if its stiffness ratio  $R_{\text{stiff}}$  is large. The stiffness ratio  $R_{\text{stiff}}$  characterizes the degree of stiffness. It needs to be pointed out that this is a heuristic measure of stiffness: A large  $R_{\text{stiff}}$  may (often does) lead to a stiff differential equation, but this is not guaranteed. A linear system can be constructed with a large  $R_{\text{stiff}}$  (e.g., due to scaling) but is not stiff [18].

Such a definition links stiffness to the condition number of the matrix  $F$ : A large  $R_{\text{stiff}}$  indicates the condition number of  $F$  is bad (or large) and vice versa [10]. Adopting such definition of stiffness ratio, we next examine the stiffness

of the stochastic differential equation (4) by considering the condition number of its Jacobian matrix

$$F(Q, \lambda) = \nabla_\alpha f^T|_{\alpha=0} = A(\lambda) - QB(\lambda), \quad (12)$$

where

$$A(\lambda) = \nabla_\alpha f_1^T|_{\alpha=0} = -\frac{1}{2}(\nabla_x \nabla_x^T \log p)^{-1} \nabla_x \nabla_x^T \log h|_{x=0}, \quad (13)$$

$$B(\lambda) = -\nabla_\alpha f_2^T|_{\alpha=0} = -\frac{1}{2} \nabla_x \nabla_x^T \log p|_{x=0}. \quad (14)$$

We expressly denote  $F$  as a function of  $Q$ . Our goal is to choose  $Q$  to reduce the stiffness of  $F(Q, \lambda)$  by reducing its condition number. We limit  $Q$  as a function of  $\lambda$  only and it is independent of  $\alpha$ .

For the matrix  $F(Q, \lambda)$  in (12), its condition number is defined as

$$\kappa_\nu(F) = \begin{cases} \|F\|_\nu \|F^{-1}\|_\nu, & \text{if } F \text{ is nonsingular,} \\ \infty, & \text{if } F \text{ is singular,} \end{cases}$$

in which  $\|\cdot\|_\nu$  is a matrix norm. The value of a condition number is dependent of the matrix norm. We always have  $\kappa_\nu(F) \geq 1$ . For any scalar  $c \neq 0$ , the condition number of  $F = cI_n$  is  $\kappa_\nu(cI_n) = 1$ .

**THEOREM 3.1.** Assume that  $F(Q, \lambda)$  exists and is finite, and that  $B(\lambda)$  is positive definite for all  $\lambda \in [0, 1]$ . Then for any matrix norm  $\|\cdot\|_\nu$ , we have

$$\lim_{c \rightarrow +\infty} \min_{Q \geq 0, \|Q\|_\nu = c} \kappa_\nu(F) = 1. \quad (15)$$

**PROOF.** Under the assumption that  $B$  is positive definite for all  $\lambda \in [0, 1]$ , choose  $Q = cB^{-1}, c > 0$ . Then  $Q$  is positive definite and

$$\begin{aligned} \kappa_\nu(F) &= \kappa_\nu(A - cI_n) \\ &= \|I_n - c^{-1}A\|_\nu \|(I_n - c^{-1}A)^{-1}\|_\nu \\ &\rightarrow \|I_n\|_\nu \|(I_n)^{-1}\|_\nu = 1, \end{aligned}$$

as  $c \rightarrow +\infty$ . Q.E.D.

The assumption that  $B$  is positive definite is mild and satisfied by unimodal densities that are log-concave. Therefore, under mild conditions, we can always find a  $Q \geq 0$  sufficiently large so that the flow equation (4) is no longer considered stiff. However, other considerations need to be included in the choice of  $Q$ . The continuous flow equation (4) needs to be discretized for numerical evaluation in applications, which would introduce numerical errors. There are many methods for numerical evaluation of continuous stochastic differential equations [16]. As the first step, we consider the Euler-Maruyama method as the following

$$\begin{aligned} \alpha_k &= \alpha_{k-1} + [f_1(\alpha_{k-1}, \lambda_{k-1}) + Qf_2(\alpha_{k-1}, \lambda_{k-1})]\Delta\lambda \\ &\quad + q(\lambda_{k-1})dw_k, k = 1, 2, \dots, K, \\ \alpha_0 &= \alpha(0), \end{aligned} \quad (16)$$

in which  $dw_k = w_{\lambda_k} - w_{\lambda_{k-1}}$ ,  $\Delta\lambda$  is the step size,  $\lambda_k = k\Delta\lambda$ , and  $K$  is the total number of steps. Without loss of generality, we assume that  $\lambda_K = 1$ , i.e.,  $\Delta\lambda = 1/K$ .

Define a stochastic process  $\{\tilde{\alpha}(\lambda), \lambda \in [0, 1]\}$  continuous in  $\lambda$  as the following

$$\begin{aligned}\tilde{\alpha}(\lambda) &= \alpha_{\lfloor \lambda/\Delta\lambda \rfloor} + \int_{\lfloor \lambda/\Delta\lambda \rfloor \Delta\lambda}^{\lambda} [f^s(\alpha_{\lfloor s/\Delta\lambda \rfloor}, \lfloor s/\Delta\lambda \rfloor \Delta\lambda) ds \\ &\quad + q(\lfloor s/\Delta\lambda \rfloor \Delta\lambda) dw_s], \\ \tilde{\alpha}(0) &= \alpha(0),\end{aligned}$$

where  $\lfloor \cdot \rfloor$  is the floor function. Consider the numerical error

$$e(\lambda) = \alpha(\lambda) - \tilde{\alpha}(\lambda), \lambda \in [0, 1].$$

Direct analysis of the error  $e(\lambda)$  as a functional of  $Q$  is generally challenging. In the next section, we use a simple example to numerically illustrate the dependence of  $e(\lambda)$  on  $Q$ .

We next examine the relationship between the covariance matrix of  $\alpha_k$  and  $Q$ . Consider a special case of linear flow (4) in which  $f_1$  and  $f_2$  in (7) are linear in  $\alpha$ . For this case, using the notations (13)-(14), we may write

$$f_1 = A(\lambda)\alpha + b_1(\lambda), \quad f_2 = -B(\lambda)\alpha - b_2(\lambda)$$

in which  $b_1(\lambda), b_2(\lambda) \in \mathbb{R}^n$  are  $n$ -dimensional vector functions of  $\lambda$  only. Consequently, (16) becomes

$$\begin{aligned}\alpha_k &= [I_n + (A_{k-1} - Q_{k-1}B_{k-1})\Delta\lambda]\alpha_{k-1} \\ &\quad + [b_1(\lambda_{k-1}) - Q_{k-1}b_2(\lambda_{k-1})]\Delta\lambda + q(\lambda_{k-1})dw_k,\end{aligned}$$

where  $A_{k-1} = A(\lambda_{k-1})$ ,  $B_{k-1} = B(\lambda_{k-1})$ , and  $Q_{k-1} = Q(\lambda_{k-1})$ . Let  $V_k$  denote the covariance matrix of  $\alpha_k$ . Then  $V_k$  satisfies

$$\begin{aligned}V_k &= [I_n + (A_{k-1} - Q_{k-1}B_{k-1})\Delta\lambda]V_{k-1}[I_n + \\ &\quad (A_{k-1} - Q_{k-1}B_{k-1})\Delta\lambda]^T + Q_{k-1}\Delta\lambda.\end{aligned}\quad (17)$$

Starting from  $k = 1$ , we may recursively derive  $V_k$  to obtain

$$V_k = \Phi(k)V_0\Phi^T(k) + \Delta\lambda \sum_{i=1}^k \Phi(i-1)Q_{k-i}\Phi^T(i-1) \quad (18)$$

for  $k = 1, 2, \dots, K$ , where  $\Phi(0) = I_n$  and

$$\Phi(i) = \prod_{j=1}^i [I_n + (A_{j-1} - Q_{j-1}B_{j-1})\Delta\lambda], i \geq 1.$$

By construction, the density of  $\alpha(\lambda)$  is  $p(x, \lambda)$  for any  $Q$  positive semi-definite. Consequently, the variance of  $\alpha(\lambda)$  is not dependent of  $Q$ . The matrix  $B$  is typically positive definite. Assume  $Q$  is a constant,  $Q_k = Q, k = 1, 2, \dots, K$ . For sufficiently large  $Q$ , the variance of the discrete approximation  $\alpha_k$  is dependent of  $Q$  and, for given  $\Delta\lambda$ , grows exponentially in  $Q$  according to (18).

Eqn. (18) also reveals that for a given  $Q$ , we may reduce  $\Delta\lambda$  to achieve desired accuracy, but that would be done at the cost of increased computation for each particle. The

increase is exponentially proportional to  $\|Q\|_2$  according to (18). Therefore,  $\|Q\|_2$  should be as small as possible to reduce the error in the numerical evaluation of the stochastic flow (4).

To summarize, the choice of  $Q$  should balance at least the following three conflicting factors:

- (1) The value of  $\|Q\|_2$  may need to be sufficiently large to reduce the condition number of  $F(Q, \lambda)$  in (13) according to Theorem 3.1.
- (2) The value of  $\|Q\|_2$  should not be too large to reduce the variance of  $\alpha_k$  according to (18), assuming the Euler-Maruyama method (16) is used.
- (3) The Euler-Maruyama method (16) should be stable in the following sense: The linearized form of (16) at  $\alpha = 0$  is

$$\alpha_k = (I_n + F\Delta\lambda)\alpha_{k-1} + qdw_k, k = 1, 2, \dots, K. \quad (19)$$

For this linearized difference equation to be stable, we must have  $\|I_n + F\Delta\lambda\|_2 < 1$ , which also limits the value of  $Q$  for a given  $\Delta\lambda$ .

Given these conflicting factors, we need to balance reducing the condition number of  $F = A - QB$  with the norm of  $Q$ . To that end, we choose  $Q$  by minimizing an objective function  $J(Q)$ . Two choices of  $J(Q)$  are considered below.

- (1) Consider a special form of  $Q$ :  $Q = cB^{-1}$  in which scalar  $c \geq 0$  is a design parameter, and define

$$J_1(Q) = \|F\|_2 \|F^{-1}\|_2 + \mu c.$$

- (2) Consider the Frobenius norm and define

$$J_2(Q) = \|F + \gamma I_n\|_F^2 + \mu \|Q\|_F^2.$$

In  $J_1(Q)$ ,  $J_2(Q)$ ,  $\mu > 0$ ,  $\gamma > 0$  are user adjustable parameters. The objective function  $J_1(Q)$  limits the choice of  $Q$  to a specific form. For a scalar  $\gamma \neq 0$ , the condition number of the matrix  $\gamma I_n$  is always one, the minimum. The objective function  $J_2(Q)$  is to select  $Q$  such that  $F$  is as close as possible to  $-\gamma I_n$  for a positive scalar  $\gamma$ . Note that the negative sign is needed because  $F$  shall be stable, i.e., its eigenvalues have strictly negative real parts. Analytical solutions of the optimal  $Q$  can be obtained for these objective functions. The numerical example in the next section shows that their optimal solutions provide competing filtering performance.

**THEOREM 3.2.** Assume that all real parts of the eigenvalues of  $A(\lambda)$  are non-positive and  $|Re\bar{\lambda}| > 0$  for all  $\lambda \in [0, 1]$  and that  $B(\lambda)$  is positive definite for all  $\lambda \in [0, 1]$ . Then

- (a) the optimal solution  $Q^*$  that minimizes  $J_1(Q)$  is

$$Q^* = Q_c, \quad Q_c = c^* B^{-1},$$

$$c^* = \max\left\{\sqrt{\frac{|Re\bar{\lambda}| - |Re\lambda|}{\mu}} - |Re\lambda|, 0\right\}, \quad (20)$$

in which  $|Re\bar{\lambda}|$  and  $|Re\lambda|$  are defined in (11) for the matrix  $A$ .

- (b) The optimal solution  $Q^*$  that minimizes  $J_2(Q)$  is

$$Q^* = Q_\gamma, \quad Q_\gamma = \mathbb{P}^+[X], \quad (21)$$

where  $\mathbb{P}^+[\cdot]$  is matrix projection into the space of positive semi-definite matrices by replacing negative eigenvalues with 0, and  $X$  is the solution to the following Sylvester equation

$$X(B^2 + \mu I_n) + (B^2 + \mu I_n)X = AB + BA^T + 2\gamma B. \quad (22)$$

PROOF. (a) For the specific choice of  $Q = cB^{-1}$ ,  $F = A - cI$ . Therefore,

$$\kappa_2(F) = \frac{|Re\bar{\lambda}| + c}{|Re\lambda| + c}$$

which is a decreasing function of  $c > 0$ , and

$$J_1(Q) = \frac{|Re\bar{\lambda}| + c}{|Re\lambda| + c} + \mu c.$$

The optimal value of  $c$  that minimizes  $J_1(Q)$  is given by (20). (b) Note that for symmetric matrices  $Q$  and  $B$ , the objective function  $J_2(Q)$  is

$$J_2(Q) = \text{tr}((A + \gamma I_n)(A^T + \gamma I_n)) - \text{tr}(QB(A^T + \gamma I_n)) - \text{tr}((A + \gamma I_n)BQ) + \text{tr}(QB^2Q) + \mu \text{tr}(Q^2). \quad (23)$$

For any two square matrices  $X$  and  $Q$  of the same dimensions and symmetric matrix  $Q$ , the derivative of  $\text{tr}(XQ)$  with respect to  $Q$  is  $\partial \text{tr}(XQ)/\partial Q = X + X^T - X \circ I$  in which  $\circ$  is the Hadamard (elementwise) product [27]. Using this formula, and also the trace identify  $\text{tr}(QX) = \text{tr}(XQ)$ , we arrive at

$$\frac{\partial J_2(Q)}{\partial Q} = 2R - R \circ I_n$$

where

$$R = -(A + \gamma I_n)B - B(A^T + \gamma I_n) + QB^2 + B^2Q + 2\mu Q.$$

Note that  $2X - X \circ I_n = 0$  if and only if  $X = 0$ . Setting the derivative of  $J_2(Q)$  with respect  $Q$  to zero, we obtain  $R = 0$ , from which we know that  $Q$  satisfies

$$Q(B^2 + \mu I_n) + (B^2 + \mu I_n)Q = AB + BA^T + 2\gamma B$$

which is (22). The matrix projection operation  $\mathbb{P}^+[\cdot]$  is needed since the optimal  $Q^* = Q_\gamma$  must be positive semi-definite. Q.E.D.

The exact form of the solution  $X$  to the Sylvester equation (22) is given by

$$\text{vec}(X) = Y_1^{-1} \text{vec}(Y_2)$$

in which

$$Y_1 = (B^2 + \mu I_n) \otimes I_n + I_n \otimes (B^2 + \mu I_n),$$

$$Y_2 = AB + BA^T + 2\gamma B,$$

where  $\otimes$  is the Kronecker product and  $\text{vec}(\cdot)$  is the matrix vec operator.

## IV. NUMERICAL EXAMPLES

### Example 1

We first analyze the error of the Euler-Maruyama method and obtain an exact relationship between the error and the diffusion matrix  $Q$  for a simple special case.

Consider a one-dimensional linear flow (4), which is possible if  $g$  and  $h$  are Gaussian or exponential [5]. Assume that  $q \neq 0$  is a scalar constant, and denote

$$f_1 = a_1\alpha + b_1, \quad f_2 = -a_2\alpha - b_2, a_2 > 0.$$

Then the stochastic differential equation (4) becomes

$$d\alpha = (a\alpha + b)d\lambda + qdw_\lambda,$$

where  $a = a_1 - Qa_2$ ,  $b = b_1 - Qb_2$ , and  $Q = q^2$ . Assume that  $a \neq 0$  to avoid degenerate case. For a fixed  $\Delta\lambda$ , we must have  $a_1/a_2 < Q < (2 + a_1\Delta\lambda)/(a_2\Delta\lambda)$  in order for the eigenvalue of  $|1 + A\Delta\lambda| = |1 + a\Delta\lambda|$  to be less than 1 for the linearized equation (19) to be stable. The solution of (4) is

$$\alpha(\lambda) = e^{a\lambda}\alpha(0) + \frac{b}{a}(e^{a\lambda} - 1) + q \int_0^\lambda e^{a(\lambda-s)} dw_s, \lambda \in [0, 1]. \quad (24)$$

The solution corresponding to the difference equation (16) is

$$\alpha_k = (1 + \Delta\lambda a)^k \alpha(0) + \frac{b}{a}[(1 + \Delta\lambda a)^k - 1] + q \sum_{i=1}^k (1 + \Delta\lambda a)^{k-i} dw_i, k = 1, 2, \dots, K. \quad (25)$$

The error  $e(1) = \alpha(1) - \alpha_K$  is

$$e(1) = (\alpha(0) + \frac{b}{a}) [e^a - (1 + \Delta\lambda a)^K] + q \left[ \int_0^1 e^{a(1-s)} dw_s - \sum_{i=1}^K (1 + \Delta\lambda a)^{K-i} dw_i \right].$$

Its mean is

$$E[e(1)] = (\alpha(0) + \frac{b}{a}) [e^a - (1 + \Delta\lambda a)^K]$$

and, according to Itô isometry, its variance is

$$\begin{aligned} \text{var}[e(1)] &= Q \sum_{i=1}^K \int_{\lambda_{i-1}}^{\lambda_i} [e^{a(1-s)} - (1 + \Delta\lambda a)^{K-i}]^2 ds \\ &= \frac{Q}{2a} (e^{2a} - 1) - \frac{2Q}{a} (e^{a\Delta\lambda} - 1) \frac{1 - (1 + \Delta\lambda a)^K e^{K\Delta\lambda a}}{1 - (1 + \Delta\lambda a)e^{\Delta\lambda a}} \\ &\quad + Q\Delta\lambda \frac{1 - (1 + \Delta\lambda a)^{2K}}{1 - (1 + \Delta\lambda a)^2}. \end{aligned}$$

Therefore,

$$E[\|e(1)\|_2^2] = (E[e(1)])^2 + \text{var}[e(1)]. \quad (26)$$

Note that  $a = a_1 - Qa_2$  is a function of  $Q$ . Even with the analytic form (26), the dependence of  $|E[e(1)]|$ ,  $\text{var}[e(1)]$ , and  $E[\|e(1)\|_2^2]$  on  $Q$  is not readily available. For  $\alpha(0) =$

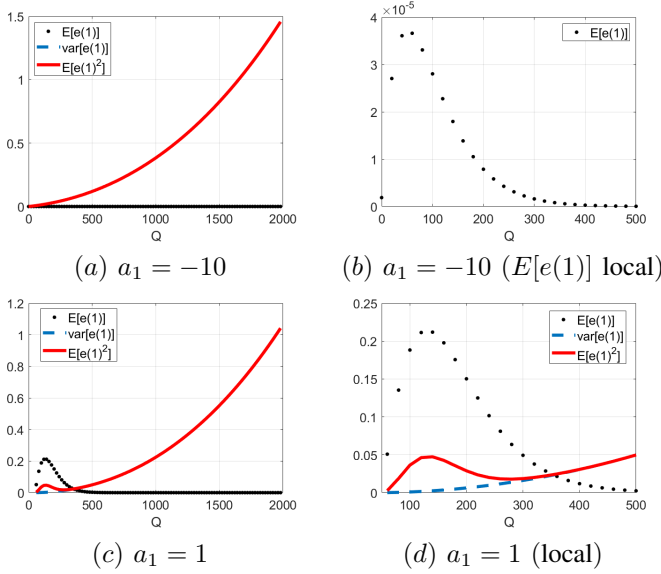


Fig. 1. The error  $e(1)$  as a function of  $Q$  for Example 1

0.1,  $b_1 = 2, b_2 = -1, a_2 = 0.02, \Delta\lambda = 0.01$ , Fig. 1 shows the dependence on  $Q$ : Fig. 1 (a) shows  $e(1)$  as a function of  $Q$  for  $a_1 = -10$ , in which  $E[\|e(1)\|_2^2]$  visually overlaps  $\text{var}[e(1)]$  because  $|E[e(1)]|$  is very small in the order of  $10^{-5}$  or smaller. Fig. 1 (b) shows  $|E[e(1)]|$  only for  $Q \in [0, 500]$  for visualization. Note that the vertical scale is at the order of  $10^{-5}$ . In Fig. 1 (c),  $a_1 = 1$ .  $|E[e(1)]|$  shows an initial increase, then decrease, and eventually  $\text{var}[e(1)]$  dominates the value of  $E[\|e(1)\|_2^2]$ . Fig. 1 (d) shows the plots for  $Q \in [0, 500]$  only for visualization. We tested  $e(1)$  for various values of parameters. Fig. 1 shows two representative cases.

For this example, numerical evaluation of  $e(1)$  shows that, for given  $\Delta\lambda$ ,

- $E[e(1)]$  may not be monotone increasing or decreasing in  $Q$  although it converges to 0 as  $Q$  goes to infinity.
- $\text{var}[e(1)]$  is exponentially increasing in  $Q$ .
- The value of  $\text{var}[e(1)]$  is an order of magnitude larger than  $E[e(1)]$ .
- There exists a  $Q_0 > 0$  such that  $E[\|e(1)\|_2^2]$  is exponentially increasing for all  $Q \geq Q_0$ .

Fig. 1 indicates that there exists an optimal  $Q$  that minimizes  $E[\|e(1)\|_2^2]$ . Further research is warranted to find such optimal  $Q$  for general case.

#### Example 2

We modified a scenario in [20] to illustrate the results. A detailed analysis of the performance of particle flow filters as compared with other popular nonlinear filters is provided in [20], [21], among other references. In this paper, our discussions are limited to evaluating the effects of the optimal diffusion matrix  $Q$  for various values of  $\mu$  and/or  $\gamma$  on filtering performance.

The scenario is illustrated in Fig. 2. The unit of position and time are meter and second, respectively. A target starts at position  $[5, 20]^T$  and moves linearly to the right with con-

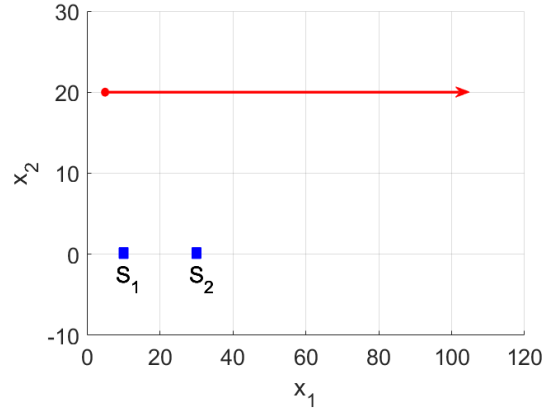


Fig. 2. A scenario modified from [20] consisting of two bearing measuring sensors  $S_1$  and  $S_2$  (blue squares). A target (red dot) starts at  $[5, 20]^T$  and moves right with constant acceleration.

stant horizontal acceleration. The state vector of the target is  $x(t) = [x_i(t)] \in \mathbb{R}^6$  in which  $[x_1(t), x_2(t)]^T$ ,  $[x_3(t), x_4(t)]^T$ , and  $[x_5(t), x_6(t)]^T$  are the location, velocity, and acceleration of the target, respectively. Target dynamics is described by the following linear stochastic differential equation

$$dx(t) = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} x(t)dt + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} dw_t, \quad (27)$$

$$x(0) = [5 \quad 20 \quad 0 \quad 0 \quad 2 \quad 0]^T,$$

in which  $w_t \in \mathbb{R}^2$  is a 2-dimensional Brownian motion in time.

There are two bearing measuring sensors: a sensor  $S_1$  located at  $s_1 = [s_1(1), s_1(2)]^T = [10, 0]^T$ , and a second sensor  $S_2$  located at  $s_2 = [s_2(1), s_2(2)]^T = [30, 0]^T$ . The nonlinear measurement equation is

$$z(t) = \begin{bmatrix} h_1(x) \\ h_2(x) \end{bmatrix} + v(t),$$

$$h_i(x) = \arctan\left(\frac{x_2 - s_i(2)}{x_1 - s_i(1)}\right), i = 1, 2,$$

in which  $v(t) \in \mathbb{R}^2$  is measurement noise. The covariance matrix of  $v_t$  is  $R = \text{diag}([0.001^2, 0.001^2])$ .

The number of particles is  $N = 500$ . Let  $\{\alpha_i(\lambda), \lambda \in [0, 1]\}$  denote the  $i$ -th particle with initial condition  $\alpha_i(0)$ ,  $i = 1, 2, \dots, N$ . For each  $i$ ,  $\alpha_i(\lambda)$  solves (7) corresponding to an initial condition  $\alpha(0)$ . For each time  $t$  when a measurement  $z(t)$  is available, the estimate of state  $\hat{x}(t)$  is updated as

$$\hat{x}(t) = \frac{1}{N} \sum_{i=1}^N \alpha_i(1).$$

The estimation error is  $e_x(t) = x(t) - \hat{x}(t)$ . Filter performance is characterized by Root Mean Square Error (RMSE) for position and velocity separately.

Of particular interest is the diffusion matrix  $Q = Q_b$ ,

$$Q_b = -(\nabla_x \nabla_x^T \log p)^{-1} (\nabla_x \nabla_x^T \log h) (\nabla_x \nabla_x^T \log p)^{-1}.$$

The matrix  $Q_b$  has been a popular choice in the design of particle flow filters in the literature [6], [7], [8], [20], [21]. The corresponding flow is called Gromov's flow. It has been observed to provide robust filtering performance in practice. We use this case as the baseline for comparison. There are two user adjustable parameters  $\mu$  and  $\gamma$  in the optimal  $Q^*$ , either  $Q_c$  or  $Q_\gamma$ . Filtering performance is evaluated by varying values of  $\mu$  and  $\gamma$ . We performed 50 Monte Carlo runs to evaluate filtering accuracy for the both baseline choice  $Q_b$  and the optimal  $Q^*$  for each choice of  $\mu$  and/or  $\gamma$ . For each run, the number of particles is 100.

To avoid overcrowded plots, two groups of parameter settings are evaluated separately. For the first group, we evaluate filtering performance corresponding to optimal  $Q_c$  by varying  $\mu$ , comparing with the baseline choice  $Q_b$ . The following five cases are considered.

- Case 1.  $Q = Q_b$ .
- Case 2.  $Q = Q_c$  with  $\mu = 0.5$ .
- Case 3.  $Q = Q_c$  with  $\mu = 1$ .
- Case 4.  $Q = Q_c$  with  $\mu = 10$ .
- Case 5.  $Q = Q_c$  with  $\mu = 20$ .

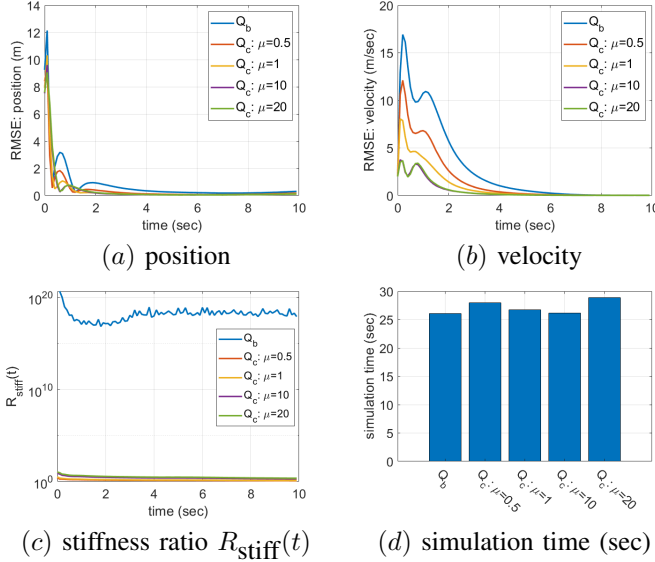


Fig. 3. Comparison of optimal  $Q_c$  with baseline choice for Example 2: Cases 1-5

RMSEs of position and velocity are shown in Fig. 3 (a)-(b), respectively. The optimal  $Q^*$  indeed leads to improvements in filtering accuracy in terms of reduced RMSE, particularly for velocity. The last two Cases 4 and 5 reveal that RMSEs for position and velocity are almost identical for  $\mu \in [10, 20]$ , which indicates some robustness in the choice of  $\mu$ . Fig. 3 (c) shows reduction of stiffness ratio for various choices of  $Q^*$ .

The stiffness ratio corresponding to  $Q = Q_b$  is theoretically infinity, but is shown as very large numbers due to numerical inaccuracy in the computation of  $Q_b$ . Fig. 3 (a)-(c) also reveal that filtering accuracy is affected by stiffness ratio as well as other factors to be discovered: For Cases 4 and 5, RMSEs for both position and velocity using  $J_1(Q)$  are lower than those for Cases 2 and 3 although the corresponding stiffness ratios are higher. Nevertheless, reducing stiffness ratios indeed leads to improved filtering performance in terms of reduced RMSEs. Fig. 3 (d) shows computing time needed to calculate the optimal  $Q_c$ , which indicates that the extra time needed to calculate the optimal value is small: The computing time for Case 4 (with  $\mu = 10$ ) is about the same as that for the baseline.

For the second group, we evaluate filtering performance corresponding to  $Q_\gamma$  by varying  $\mu$  and  $\gamma$ , comparing with the baseline choice  $Q_b$ . The following six cases are considered.

- Case 1.  $Q = Q_b$ .
- Case 6.  $Q = Q_\gamma$  with  $\mu = 10^2, \gamma = 0.2$ .
- Case 7.  $Q = Q_\gamma$  with  $\mu = 500, \gamma = 0.2$ .
- Case 8.  $Q = Q_\gamma$  with  $\mu = 10^3, \gamma = 0.2$ .
- Case 9.  $Q = Q_\gamma$  with  $\mu = 10^3, \gamma = 1$ .
- Case 10.  $Q = Q_\gamma$  with  $\mu = 10^3, \gamma = 10$ .

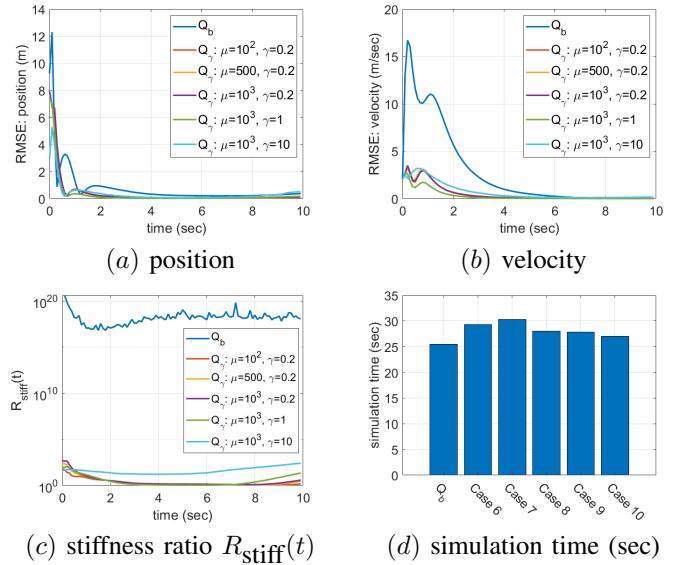


Fig. 4. Comparison of optimal  $Q$  with baseline choice for Example 2: Cases 6-10

RMSEs of position and velocity are shown in Fig. 4 (a)-(b), respectively. The optimal  $Q^*$  again leads to improvements in filtering accuracy in terms of reduced RMSE, particularly for velocity. Fig. 4 (a)-(b) show that reductions of RMSE in position and velocity are about the same for Cases 6-10, which suggests some robustness in parameter choice for this technique. Fig. 4 (d) shows that computing time needed to calculate the optimal  $Q_\gamma$  by solving the Sylvester equation (22) is between 6% and 19% of that for the baseline. Note that such extra time is fixed and not affected by the number of particles. For Case 10, the value of  $\gamma = 10$  resulted in

higher stiffness ratio, which leads to slightly larger RMSE in position and velocity. For Cases 6-10, RMSEs of the position and the velocity are comparable or smaller than those for Cases 4-5 although the corresponding stiffness ratios are higher, which again indicates that filtering accuracy is affected by stiffness ratio as well as other factors to be discovered. Accurately evaluating a continuous differential equation could be extremely challenging. Stiffness mitigation is a heuristic approach to improving filtering accuracy. Further research is needed to determine the exact dependency of filtering accuracy on the diffusion matrix  $Q$ .

Fig. 3 and Fig. 4 show that, with proper choices of  $\mu$  and/or  $\gamma$ , optimal  $Q_c$  and  $Q_\gamma$  lead to competitive filter performance.

## V. CONCLUDING REMARKS

In this paper, we investigated the role of the diffusion matrix in stiffness mitigation of the underlying stochastic differential equations for particle flow filters. We discovered that the choice of  $Q$  needs to balance several conflicting factors including stiffness mitigation and the variance of the discretized particles in numerical realization. We proposed a couple of objective functions for finding the optimal  $Q^*$  for stiffness mitigation, which in turn leads to improved filtering performance in terms of reduced filtering error. Further efforts are warranted to fully understand the exact relationship between the diffusion matrix and filtering performance in practice. In particular, it would be desirable to find the optimal diffusion matrix  $Q$  that optimizes filtering performance under certain criteria (without using the vehicle of stiffness mitigation). The link between condition number and quality of maximum likelihood covariance estimation has been reported in the literature [30]. A systematic understanding is warranted.

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